

RESEARCH METHODS

DAY 2

Welcome Back!

Day 2

**Let's start with a teamwork & a refresher.
Then we shall learn the very first step of the
research process: asking a good question!**



TEAM WORK

Let's review your homework with your teammates.
Share your curiosities and thoughts regarding your chosen global issue.

Find common themes. Identify favorite ideas. Select 1-2 questions to refine later.

A close-up photograph of a person wearing glasses, looking through a magnifying glass. The image is dimly lit, with a warm, yellowish light source visible in the background, creating a focused and investigative atmosphere.

Let's Refresh Our Memories

What is Research?



A Journey of Discovery

Research is a systematic investigation to establish facts and reach new conclusions.



Question-Driven

Every great discovery begins with a well-framed question that guides your direction.

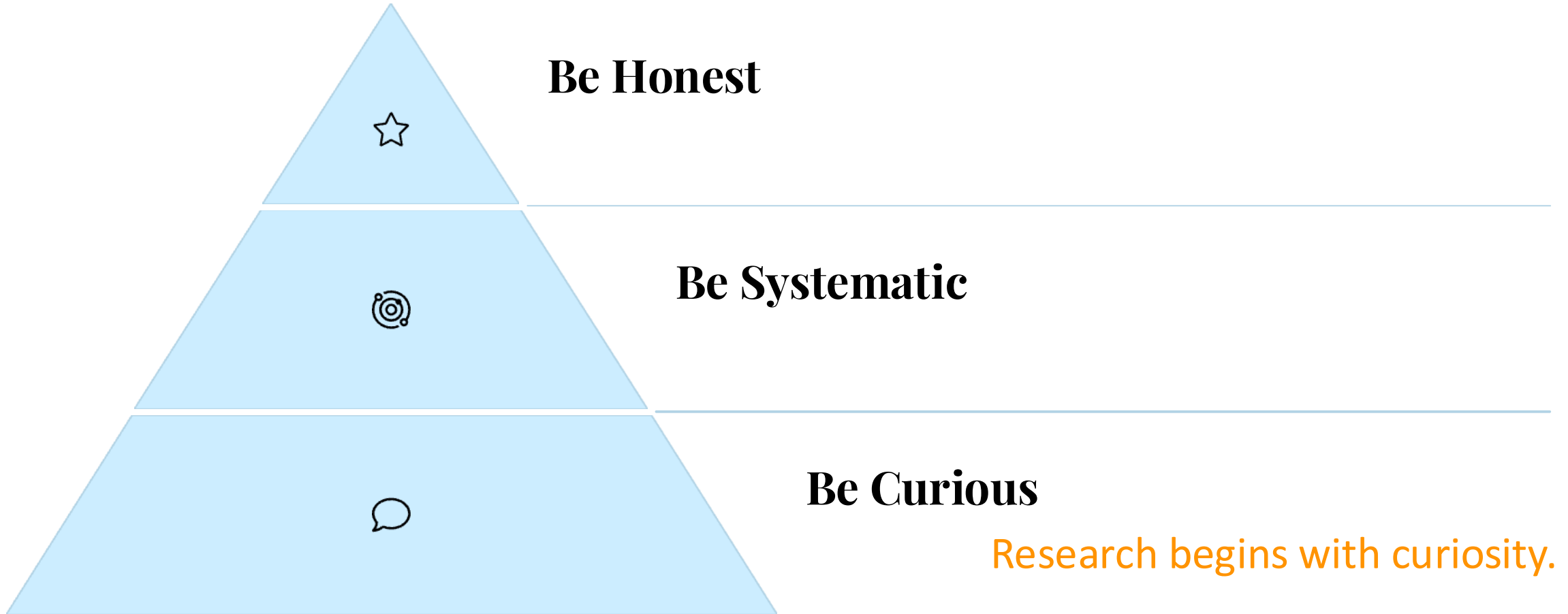


Evidence-Based

Research relies on gathering and analyzing evidence rather than opinions or assumptions.

Core Principles of Good Research

No matter what the field is



Principle 1: Be Curious

The Starting Point: Curiosity

Wonder

Asking questions such as

"Why?"	"When?"	"How much?"
"What if?"	"How come?"	"For Whom"
"Under what circumstances?"		
"What is the trend?"		
"What is the relationship between?"		

drives meaningful research.



Research begins with curiosity and asking meaningful questions.



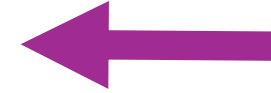
The Research Process

The journey from question to insight follows these essential steps:



Ask a Question

Formulate a clear, specific question worth investigating



Gather Information

Review the literature & collect relevant data through appropriate methods



Analyze the Information

Examine data for patterns, trends, and meaning



Draw Conclusions

Develop evidence-based answers to your original question



Share Findings

Communicate results effectively to appropriate audiences

Your Research Question is a Compass



A well-framed question acts as your compass, pointing you toward useful answers.

Vague questions lead to circular research. Specific questions drive you to meaningful discoveries.

Without the right question, even the best research efforts might not yield satisfying answers.

Case Study: The Library Problem

1

Initial Question

"How can we make teens read more books?"

2

Reframed Question

"What are the barriers to teens' library use?"

3

Research Method

Surveyed classmates about library usage

4

Discovery

Limited Wi-Fi and few comfortable spaces deterred teens



Case Study: Solar Panel Investigation



Initial Question

"How can we improve solar panel efficiency?"



Better Question

"What factors are limiting our school's solar panel performance?"



Research Findings

Shade, incorrect tilt, and dust accumulation were the issues

What Makes a Good Research Question?

BRIC-FOR

Balanced (Well-Scoped)
Relevant & **I**nteresting
Clear & **F**ocused
Open-ended
Researchable



Balanced in Scope (aka Well-Scoped Questions)

Too Broad

"What is the history of the United States?"

Spans hundreds of years and countless topics.

Too Narrow

"What was the price of bread in our town on January 1, 1950?"

More of a trivia fact than a research project.

Just Right

"How have food prices in our town changed from 1950 to today, and why?"

Focused yet allows for meaningful analysis.

Relevant and Interesting Questions

Personal Relevance

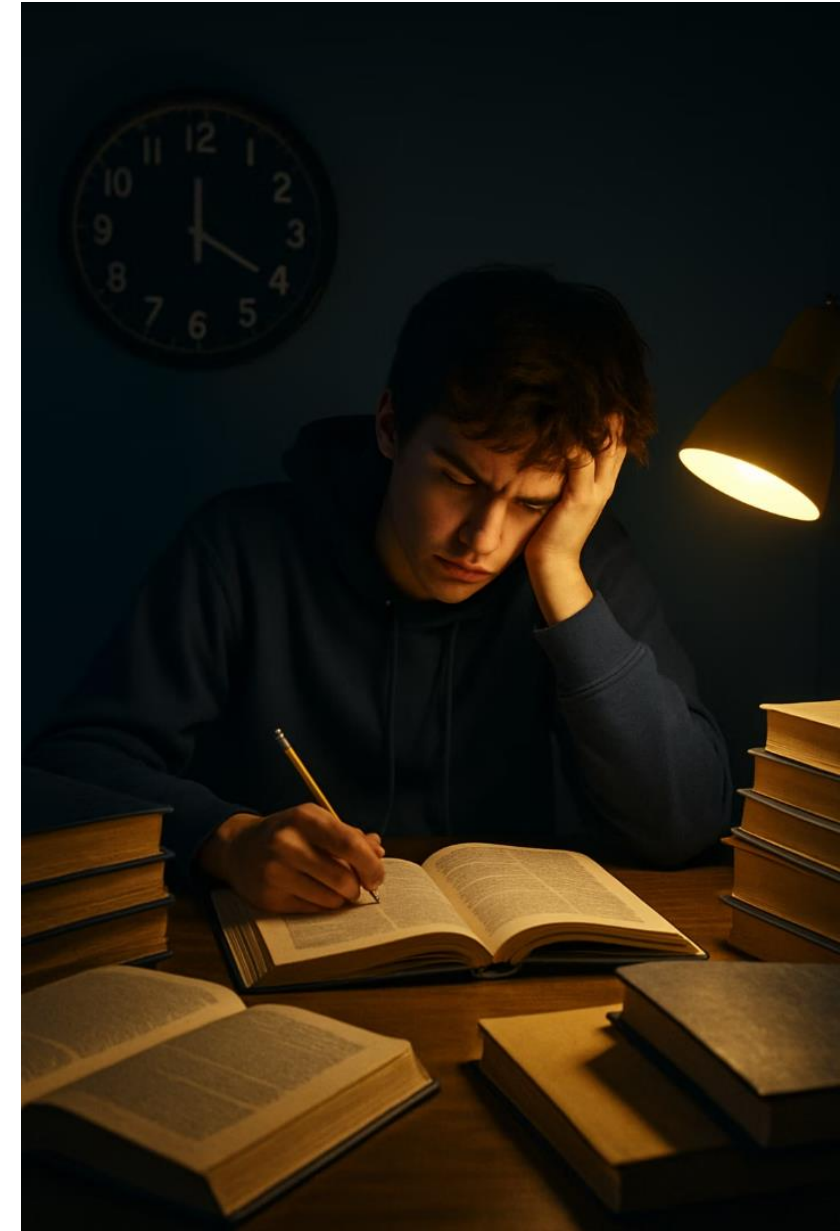
Choose topics that matter to you - you'll stay more engaged.

Community Impact

Research that addresses local issues can have real-world applications.

Example

"How does sleep affect test scores among students in my grade?"



Clear and Focused Questions

Too Broad

"What causes climate change?"

This spans science, policy, history, and economics - a lifetime of work!

Well-Focused

"How do car emissions contribute to climate change in my city?"

Narrows to one cause, one effect, and one location.

Open-Ended Questions

Closed-Ended

"Do video games affect mood?"

Can be answered with just "yes" or "no" - limited exploration.

Easy to quantify and analyze

Good for specific, categorizable information such as “Do you have a library card?”. Closed-ended questions can be used for gathering straightforward facts or opinions you can count

Open-Ended

"How might playing video games affect the mood of middle school students?"

Invites exploration of different effects, game types, and conditions.

- Require detailed, free-form responses
- Provide rich, nuanced information
- Harder to analyze quickly

Researchable Questions

Not Easily Researchable

"Is there life on other planets?"

Fascinating but beyond a student's ability to gather original data.

Researchable Version

"What do high school students believe about life on other planets?"
Can be investigated through surveys and interviews.

Key Consideration

Can you gather information to answer this using available tools?



Weak vs. Strong Research Questions

"Is pollution bad?"

Weak!

Vague, obvious, closed-ended and not specific about type of pollution or effects



Weak vs. Strong Research Questions

"What effects does air pollution have on respiratory health in children?"

Strong!

Specifies pollution type, health aspect, and affected population.



Weak vs. Strong Research Questions

"How do we fix the education system?"

Weak!

Too broad, assumes problems, and implies one solution exists.



Weak vs. Strong Research Questions

"What challenges do students face in remote learning, and how do these affect performance?"

Strong!

Focused on specific context with clear variables to examine.



Narrowing Down a Topic



Start with a broad interest

Example: "Climate change" - a general theme



Find a niche or sub-topic

Example: "Effects of rising sea levels on coastal communities"



Apply the 5 W's (and How)

Who, What, Where, When, Why, and How



Turn it into a specific question

Example: "How are rising sea levels affecting fishing communities in Florida?"

Using the 5 W's (and How)



Who

Who is involved or affected?
(teenagers, athletes, teachers,
polar bears)



What

What aspect interests you?
(mental health, economic cost,
performance)



Where

A certain location or context?
(your city, online communities,
lakes)



When

A time period? (past 5 years, during summer, historical
period)



Why/How

Causes or processes? (why something happens, how
factors interact)

Example: Narrowing "Video Games"



Broad Topic

Video games (huge topic with many aspects)



Narrower Focus

Video games and learning (educational aspect)



Specific Angle

Video games and problem-solving skills

A background image showing three students in a classroom setting, looking at a screen and taking notes.

From Topic to Research Question

Broad Topic

Video games

Narrower Focus

Video games and problem-solving skills

Research Question

"Can playing video games improve middle school students' problem-solving skills, and if so, how?"

Elements

- Who: middle school students
- What: effect on problem-solving skills
- Open-ended: asks "how" not just "if"

YOUR TURN! Let's Team UP to merge & refine your po

Make sure to use the 5 W's (& H)



Who

Who is involved or affected?
(teenagers, athletes, teachers, polar bears)



What

What aspect interests you? (mental health, economic cost, performance)



Where

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A time period? (past 5 years, during summer, historical period)



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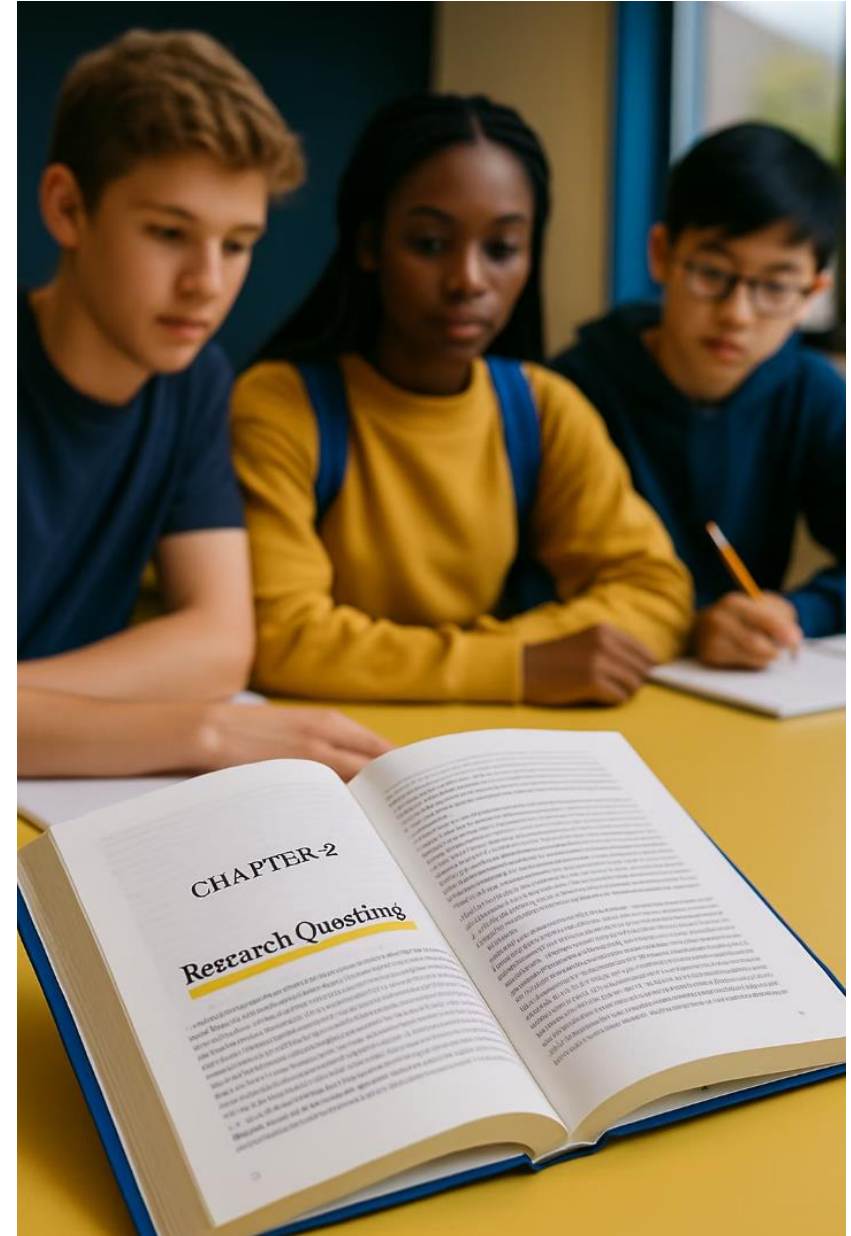
Team Homework: Handout 2

From curiosity to formulating research questions

What you know/assume & want to learn

3-5 narrowed down research questions

Evaluate them using BRIC-FOR



Homework

WHAT WE KNOW ABOUT THIS ISSUE	WHAT WE WANT TO KNOW

Homework

Come up with 3 - 5 potential research questions

**RATE each question using
BRIC-FOR criteria**

Balanced spectrum? Relevant? Interesting? Clear? Focused? Open-ended? Researchable?

1

2

3

4

5

Not so much

Somewhat

Very much

EXTRA SLIDES FOR DAY 2

Let's Play a Game!

Let's take your research questions to the next level.

Each team will receive a card with a “constraint”.

Update your question following the hint on the card.

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Update your question following the hint on the card.

**Now it is time present your question to the larger group and
get rated on BRIC-FOR**

Balanced spectrum? Relevant? Interesting? Clear? Focused? Open-ended? Researchable?

1	2	3	4	5
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Not so much

Somewhat

Very much